



CUBAN NATIONAL CHEMISTRY CONTEST 2019

"Imagination is more important than knowledge."

Albert Einstein

Name(s): _____ Last name(s): _____
School: _____
Municipality: _____ Province: _____
CODE: _____ Test: **DAY 2. ELEVENTH GRADE EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 2 problem set, with an individual exam for 11th grade. The exam has 3 problems (problem 7, 8 items; problems 8, 8 items; problems 9, 7 items) and a total of 6 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), and each item scoring (in marks) is indicated.

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Equation of ideal gas	$p \cdot V = n \cdot R \cdot T$
Gases constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta E + W$
Zero on Celsius scale	273,15 K	Gibbs' free energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

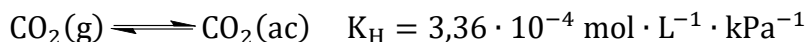
PERIODIC TABLE OF ELEMENTS

1 1A																		18 8A	
1 H 1.008	2 2A																2 He 4.003		
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18		
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95		
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80		
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3		
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)		
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)		
58 Ce 140.1		59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0					
90 Th 232.0		91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)					

Problem 7. Solutions (15 points)

7.1	7.2	7.3	7.4	7.5	7.6	7.7	7.8	Total	Total
5	2	2	4	5	3	5	4	30	15

Henry's constant, K_H , relates the molar concentration with the partial pressure of the gas saturating the solvent at a given temperature. For example, at 25 °C for carbon dioxide,



The solubility of gasses can also be expressed in terms of the solubility coefficient α , defined as the volume of the gas (in milliliters) measured at standard conditions (0 °C and 101,3 kPa) that dissolves in 1 mL of water when the partial pressure of the gas is 101,3 kPa.

7.1. Find an approximate expression relating K_H and α . Consider that the volume of the solution does not depend on the amount of dissolved gas.

7.2. How do you explain the difference between the solubilities of gases SO_2 , H_2S and O_2 ?

α (at 25 °C): $\text{SO}_2 = 32,79$; $\text{H}_2\text{S} = 2,282$; $\text{O}_2 = 0,02831$.

7.3. Why the solubility of gasses decreases when the temperature raises?

Hydrogen chloride is a gas very soluble in water. At temperatures below 0 °C, it is possible to obtain hydrates of well-defined composition $\text{HCl} \cdot n\text{H}_2\text{O}$, where n is an integer number.

7.4. Find the formula of a hydrogen chloride hydrate that melts at -70 °C knowing that it contains 9,1% mass of hydrogen.

For solid substances, the solubility coefficient S is commonly used. It is defined as the solute mass that dissolves in 100 g of water. In addition, for sparingly soluble ionic salutes there is a direct relation between the coefficient S and the solubility product K_{ps} .

7.5. Demonstrate that for PbCl_2 the following relation exists, between the solubility coefficient S and the constant K_{ps} . Consider that the density of pure water is $1,00 \text{ g} \cdot \text{mL}^{-1}$ at 25 °C.

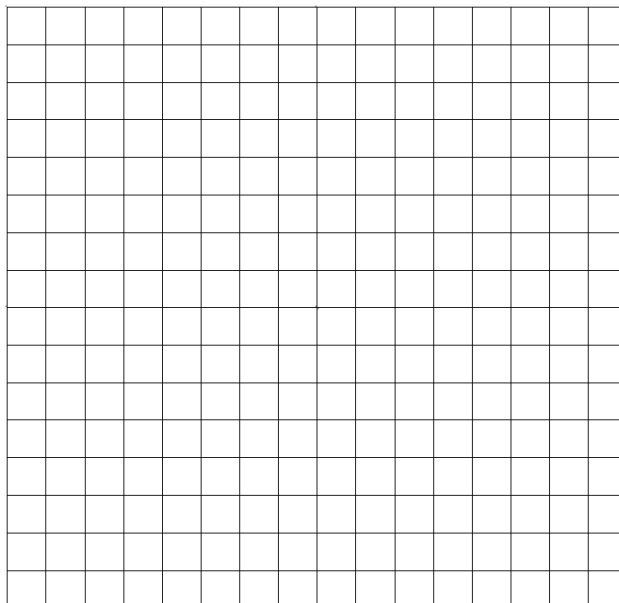
$$K_{ps} \approx 4000 \cdot \frac{S^3}{M^3}$$

A student wanted to indirectly calculate the standard enthalpy variation of the dissolution of lead(II) chloride. To attain this objective, the student determined the solubility coefficient S of PbCl_2 at several temperatures. The results obtained by the student are shown in the following.

T (°C)	0	10	20	30	40
1/T (K ⁻¹)					
S	0,67	0,82	1,00	1,20	1,42
K_{ps}					
$\ln K_{ps}$					

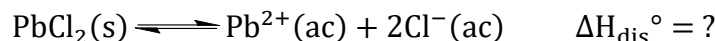
7.6. Complete the missing information in the previous table. Suppose that the expression from item 7.5 is valid for all the temperatures. $M(\text{PbCl}_2) = 278 \text{ g} \cdot \text{mL}^{-1}$.

7.7. Draw a plot of $\ln K_{ps}$ vs $1/T$ in the indicated space below. Use an adequate scale.



7.8. Calculate the value of the standard enthalpy variation for the dissolution of PbCl_2 . Use the plot elaborated by you in item 7.7 and the following relation.

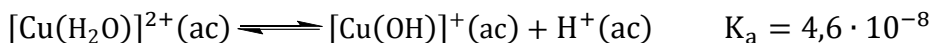
$$\ln K_{ps} = -\frac{\Delta H_{dis}^{\circ}}{R} \cdot \frac{1}{T} + \frac{\Delta S_{dis}^{\circ}}{R}$$



Problem 8. Ionic equilibrium (20 points)

8.1	8.2	8.3	8.4	8.5	8.6	8.7	8.8	Total	Total
3	2	4	6	3	8	2	12	40	20

The solutions of salts containing copper(II) ions are slightly acid, because this ion is hydrated and behaves as a weak acid in its hydrolysis. The process can be simplified to

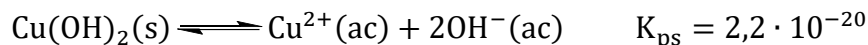


8.1. Calculate the pH of an aqueous solution of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ $0,50 \text{ mol} \cdot \text{L}^{-1}$.

When sodium hydroxide is added to a solution of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, the formation of a light blue colored solid characteristic of copper(II) hydroxide.

8.2. Which condition is necessary for a precipitate to form?

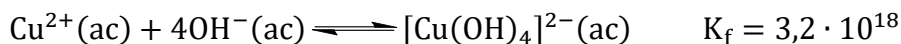
8.3. Calculate the solubility (in $\text{mol} \cdot \text{L}^{-1}$) of copper(II) hydroxide in pure water.



8.4. 25 mL of an aqueous solution of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ $0,50 \text{ mol} \cdot \text{L}^{-1}$ were mixed with 25 mL of an aqueous solution of NaOH $1,0 \text{ mol} \cdot \text{L}^{-1}$. Calculate the concentration of ions Cu^{2+} present in the resultant solution when the equilibrium is reached.

If the addition of strong base continues, the precipitate dissolves originating a marine-blue colored solution, due to the formation of the ion tetrahydrodecuprate(II), $[\text{Cu}(\text{OH})_4]^{-2}$.

8.5. Given the value of the formation constant K_f of tetrahydroxidecuprate(II) from ions copper(II) and hydroxide, calculate the value of the equilibrium constant K between the hydroxide and complex ions.



8.6. 25 mL of an aqueous solution of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ $0,50 \text{ mol} \cdot \text{L}^{-1}$ were mixed with 75 mL of an aqueous solution of NaOH $1,0 \text{ mol} \cdot \text{L}^{-1}$. Calculate the concentration of Cu^{2+} ions present in the resultant solution when the equilibrium is reached.

8.7. Classify copper(II) hydroxide according to the acid-base criteria.

8.8. 25 mL of an aqueous solution of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ $0,50 \text{ mol} \cdot \text{L}^{-1}$ were mixed with 75 mL of an aqueous solution of NaCN $1,0 \text{ mol} \cdot \text{L}^{-1}$. Calculate the concentration of Cu^{2+} ions in the resultant solution when the equilibrium was reached.



Problem 9. Electrochemistry (15 points)

9.1	9.2	9.3	9.4	9.5	9.6	9.7	Total	Total
7	3	4	4	3	3	6	30	15

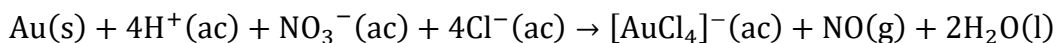
Copper, silver and gold are transition elements belonging to the 11th group in the periodic table. Their corresponding simple substances are metals and are employed since the antiquity for the manufacture of coins.

Highly pure copper (higher than 99,99%) is obtained by electrodeposition in a copper cathode. In the electrolytic cell impure copper is placed in the anode and in the cathode a plate made of pure copper is placed. An aqueous solution of copper(II) sulfate is employed as electrolytic conductor.

9.1. Suppose that in the electrolytic cell described above a copper anode containing impurities of gold, platinum and silver is placed. Write all the possible anodic and cathodic half reactions and select which ones are thermodynamically more favored. Assume standard conditions.

9.2. A copper anode has approximate dimensions of $100 \times 125 \text{ cm}$, a thick of 5 cm and an approximate weight of 350 kg. Calculate the minimum amount of current (in Coulomb) to carry out the electrorefining of a copper anode with an approximate purity of 96%.

Copper, silver and gold are not susceptible to oxidation by H^{+} ions in standard conditions and therefore they are employed, together with platinum, in ornaments and jewelry. Nitric acid is able to dissolve copper and silver; while aqua regia, a 3:1 mixture of concentrated hydrochloric and nitric acids, is an effective reagent for the oxidation of gold and platinum. In aqua regia, NO_3^{-} ions provide the oxidizing power and Cl^{-} ions act as complexing agents. The total reaction occurring with gold is:



9.3. Demonstrate that the oxidation of copper and silver with nitric acid is spontaneous.

9.4. Write half-equations for the oxidation and reduction corresponding to the dissolution of gold with aqua regia.

It is believed that the active oxidizing species in the aqua regia solution are chlorine and nitrosyl chloride, generated by the reaction

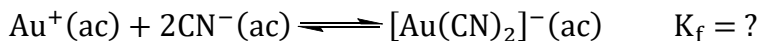


9.5. Represent the structure of NOCl molecule.

Gold is extracted by lixiviation with cyanide. Using facilitates gold oxidation through the formation of complex ion $[\text{Au}(\text{CN})_2]^-$. To recover the dissolved gold a reductant reagent is employed, for example, zinc dust. Today exist several communities all around the globe that fight against the negative impacts in economy, society and environment caused by the mining activity for obtaining gold using cyanide.

9.6. Write the chemical equation corresponding to the reduction of gold complex ion with zinc.

9.7. Calculate the constant of formation K_f of complex ion $[\text{Au}(\text{CN})_2]^-$ from ions Au^+ and CN^- with the following data.



$E^\circ(\text{V})$ para $\text{pH} = 0$:

$\text{Cu}^{2+}(\text{ac})/\text{Cu}(\text{s}) = 0,340$; $\text{Au}^{3+}(\text{ac})/\text{Au}(\text{s}) = 1,52$; $\text{Pt}^{2+}(\text{ac})/\text{Pt}(\text{s}) = 1,188$; $\text{Ag}^+(\text{ac})/\text{Ag}(\text{s}) = 0,799$;
 $\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{l}) = 1,229$; $\text{H}^+(\text{ac})/\text{H}_2(\text{g}) = 0,00$; $\text{S}_2\text{O}_8^{2-}(\text{ac})/\text{SO}_4^{2-}(\text{ac}) = 1,96$; $\text{SO}_4^{2-}(\text{ac})/\text{SO}_2(\text{g}) = 0,158$;
 $\text{NO}_3^-(\text{ac})/\text{NO}(\text{g}) = 0,96$; $[\text{Au}(\text{CN})_2]^- (\text{ac})/\text{Au}(\text{s}) = -0,596$; $\text{Au}^+(\text{ac})/\text{Au}(\text{s}) = 1,83$.

